



Data Mining & Big Data Analytics

Part 2: Data Wrangling

Layout

- **Introduction**
- **Data Cleaning**
- **Data Transformation**
- **Data Integration**

Learning

Objectives

Learning Objectives

- Outline the processes of data wrangling, focusing on their role in preparing data for analysis and storytelling.
- Apply strategies for addressing missing data within a dataset.
- Find and address anomalous values that may distort data analysis.
- Transform quantitative and categorical variables to enhance data structure and analysis.
- Reshape data into a tidy format, ensuring that each column represents a variable and each row an observation.
- Describe approaches to combining multiple datasets for comprehensive data analysis.



Section 1

Introduction

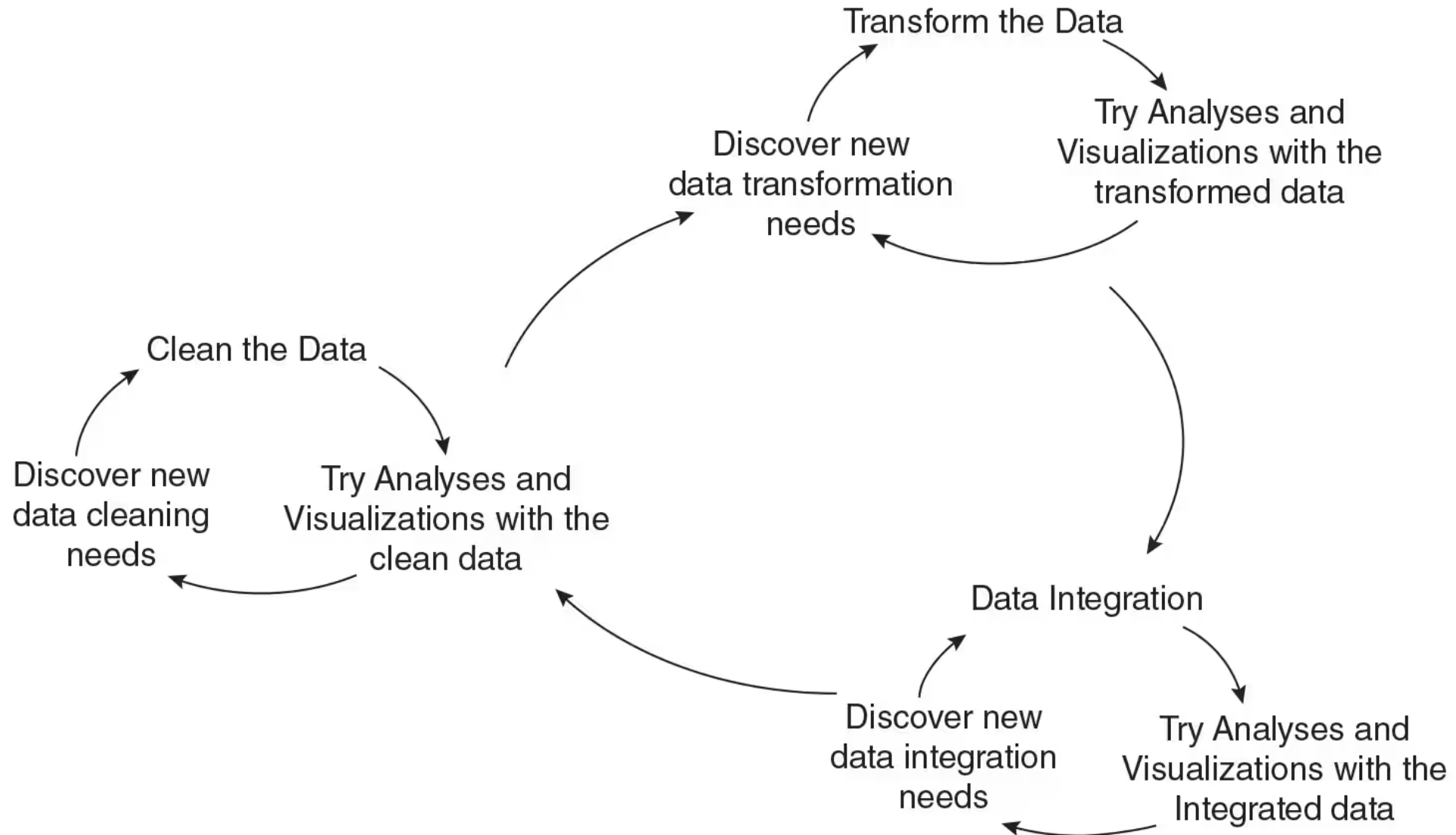
Data Wrangling Definition

- **Data wrangling**, also known as **data preprocessing**, is the process of **converting raw data into a more usable format** by **cleaning**, **transforming**, and **integrating** data to facilitate effective analysis and storytelling.

Data Wrangling Steps

- **Cleaning data** involves fixing or removing incomplete, abnormal, or inconsistent data to enhance their quality for accurate analysis and storytelling.
- **Data transformation** is the process used to convert a clean dataset into the form or structure needed for analysis and visualization.
- **Integrating data** is the process of combining multiple datasets from one or more sources to maximize the amount of information from which to draw insight.

Data Wrangling Cycle



Section 2

Data Cleaning

Data Cleaning

Main data cleaning issues:

- **Missing data**
- **Extreme data values**
- **Implausible values**
- **Incorrect formats**
- **Duplicate records**



Missing Data

Addressing missing values in data:

- Removing observations with missing values.
- Imputation from internal data.
- Imputation from external data.
- Other options.

Imputation is the process of replacing missing data with substituted values derived from patterns in available data.

first_name	age	gpa	favorite_pet	voted
Selma	18	3.1	Dog	0
Juan		3.2	Cat	1
Rozzlin	19	3.9		0



Imputation from Internal Data

Imputation means estimating the missing value based on other similar observations **or** replacing the missing value with a value that we hope is likely to be reasonably accurate.

Main common internal data imputation:

- The **mean** of a variable is the average of all the data values in a dataset for that variable.
 - It is calculated by adding up all the values and then dividing this sum by the number of values.
- The **median** of a variable is the middle value in a dataset for that variable when the values are arranged in ascending or descending order.
 - If there is an even number of values, the median is the average of the two middle values.



Imputation from Internal Data

object_id	type	material	depth	age
O1	Pottery	Clay	1	2000
O2	Jewelry	Gold	2	
O3	Tool	Bronze	1.5	3000
O4	Weapon	Iron	3.5	1500
O5	Coin	Silver	1.5	
O6	Pottery	Clay	2.5	2500
O7	Jewelry	Gold	1	
O8	Weapon	Bronze	3	2000
O9	Tool	Iron	4	
O10	Coin	Silver	1.2	1900
O11	Jewelry	Gold	2.7	
O12	Pottery	Clay	2.2	2400

object_id	type	material	depth	age
O1	Pottery	Clay	1	2000
O2	Jewelry	Gold	2	2185.71
O3	Tool	Bronze	1.5	3000
O4	Weapon	Iron	3.5	1500
O5	Coin	Silver	1.5	2185.71
O6	Pottery	Clay	2.5	2500
O7	Jewelry	Gold	1	2185.71
O8	Weapon	Bronze	3	2000
O9	Tool	Iron	4	2185.71
O10	Coin	Silver	1.2	1900
O11	Jewelry	Gold	2.7	2185.71
O12	Pottery	Clay	2.2	2400

Question

Which imputation method keep the mean value unchanged?

- Removing observations with missing values. ✓
- Imputation using the mean. ✓
- Imputation using the median. ✗



Imputation from External Data

- Imputation from external data based on values from a different dataset.
- For example, in the case of previous dataset,
 - Access the college's roster that includes the year of birth for each student.
 - Use public data about the average age of students in the class.
 - Have information about students' age from another dataset.
- However, some of them is not referred as an imputation because the values would be known, not estimated.
- We should trust that the other sources of data are reputable and relevant to our original dataset.

Other Options for Missing Data

- Replace missing values with an entirely new value specific to that missingness.
 - "NA"
 - Nonsensical value like -999.
 - For categorical variables, replace with "Missing" or "Unknown".
- Continue to include incomplete observations in the dataset, and the analysis can distinguish them from other observations where appropriate.



Other Options for Missing Data

- We could also simply pursue two separate analyses with our data:
 - one with the incomplete observations included, and
 - one with the incomplete observations removed.
- Regardless, we should always think carefully about how we modify raw data, clearly noting the choices made to deal with missing values when writing a report.



Data Missing Considerations

A data scientist must think carefully about the following things:

- What are the possible sources of missing data in my dataset?
- Is the missingness random, or do any patterns explain why values are missing?
- What strategies can I pursue to adjust my work for the missing values in my dataset?
- How can I document and communicate these missing data characteristics in my work most effectively?



Cleaning Anomalous Values

- **Implausible values** are data points that fall outside the realm of possible values, clearly indicating errors.
- **Extreme values** are data points that are noticeably different from others, potentially affecting the overall analysis in noticeable ways.
- **Incorrect formats** refer to data entries in an unexpected format, requiring correction to match the anticipated data type.
- **Duplicate records** are repeated entries (e.g., observations) in a dataset, often identified and resolved using unique identifiers.

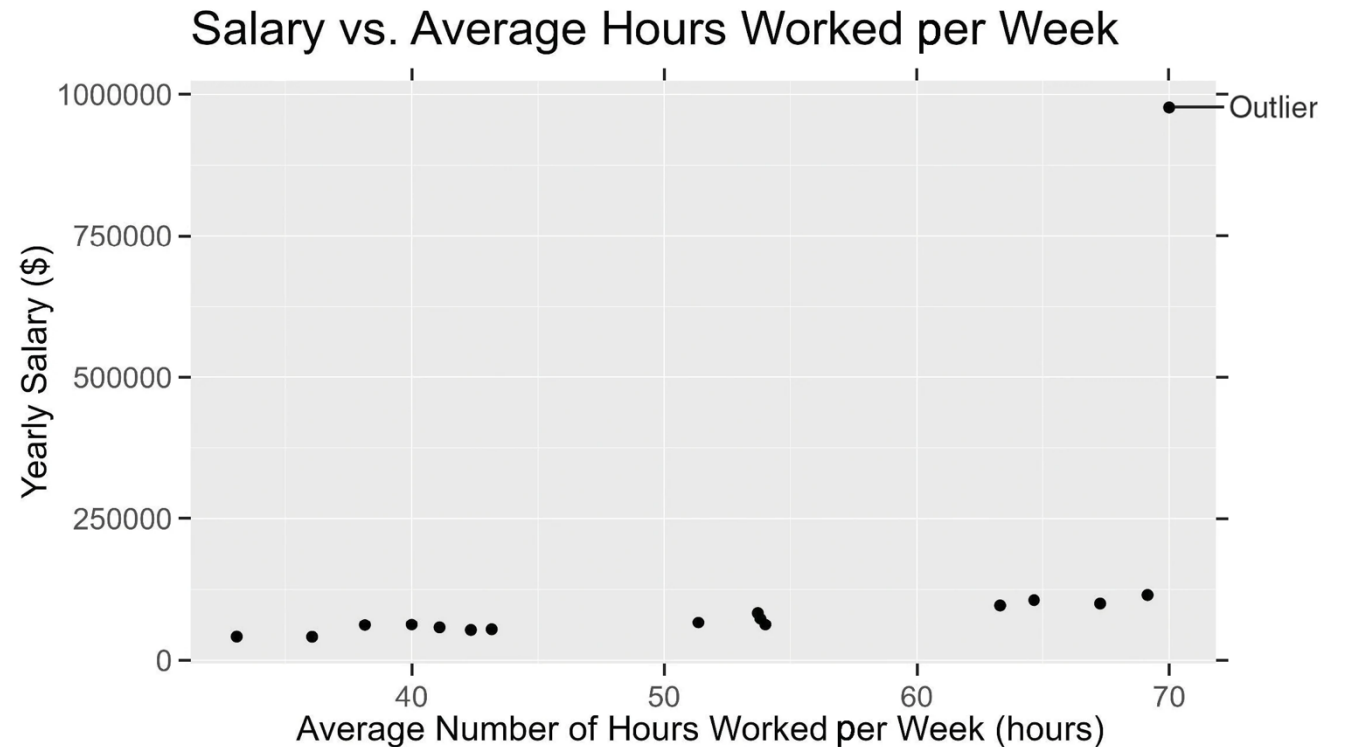
Implausible Values

- Implausible values are quantitative data that are not within a range of anticipated values, to the point of being impossible.
- For example, a customer's credit score is typically between 300 and 850. If we come across a credit score value of 22,995, or -53 .
- We could replace these cells with missing values and proceed with any of the methods previously described in our discussion of missing data.
- We could also return to the source of the data and correct these values instead of replacing them with missing values.



Extreme Values

- **Extreme data** values are exceptionally higher or lower than the other values of a variable.
- An **outlier** is an extreme data point so different from the majority that it may warrant correction or removal.
- The extremity is determined by contextual judgment and verification.



Section 3

Data Transformation

Data Transformation

- Data wrangling also includes enhancing or structuring data by creating new categorical variables from quantitative ones, changing the scale of quantitative variables, and combining multiple quantitative variables.
- Quantitative variables are numerical and allow for mathematical computations; they can be discrete or continuous.
- Categorical variables are often non-numeric and represent group or category memberships.



Transforming Quantitative Variables

- Creating a Categorical Variable from a Quantitative Variable
- Changing the Scale of a Quantitative Variable
- Combining Two or More Quantitative Variables



Categorization of Quantitative Variables

- Categorization: Converting a quantitative variable (age) into a categorical variable (age_category) by assigning labels to intervals of values.
- Example of age categorization:
 - Ages 0-11 years: child
 - Ages 12-59 years: adult
 - Ages 60+ years: senior



Categorization of Quantitative Variables

- Key principles of categorization:
 - Categories must be mutually exclusive: Each value belongs to only one category.
 - Categories must be collectively exhaustive: All values are categorized.
- Loss of information: Creating categorical variables from quantitative variables results in some loss of information, which should be done thoughtfully.

Changing the Scale of a Quantitative Variable

- **Scaling** involves creating a new quantitative variable by multiplying or dividing the original variable by a certain value.
- **Transforming** refers to applying more complex functions, such as logarithms or square roots, to the original variable.
- Scaling and transforming data help make it easier to work with and interpret, especially in visualizations.
- Example of Scaling: To make large numbers easier to read in data visualizations, you can scale down an income variable by dividing it by 1000.
 - For instance, an income \$800,000 becomes \$800 (in thousands).



Combining Two or More Quantitative Variables

- **Combining variables:** Experts often combine variables to gain deeper insights from data.
- An example is computing a "per capita" variable, which means "per person."
- GDP per capita: This is calculated by dividing a country's *GDP* by its population:

$$GDP_{\text{perCapita}} = \frac{GDP}{Population}.$$

- **Example:** The United States has a higher *GDP* than France, but their $GDP_{\text{perCapita}}$ values are closer, highlighting the importance of relative measures.



Transforming Categorical Variables

- Creating a Quantitative Variable from a Categorical Variable
- Condense the Categories of a Categorical Variable
- Splitting a Variable into Multiple Variables



Quantizing Categorical Variables

- Numeric data is essential for many analysis methods and visualization techniques.
- Transforming categorical variables into quantitative ones is crucial for machine learning applications.
- This transformation allows for numerical analysis and modeling that is not possible with categorical data.



Encoding

Encoding is the process of converting a categorical variable into a quantitative one by assigning numeric codes to the values of the categorical variable.

Boolean encoding is the process of converting a categorical variable into a quantitative one by assigning a 1 or 0 to the values of the categorical variable.

Quantizing Categorical Variables

- Text data can be translated into numeric form using Boolean variables.

student_id	age	gpa	pet_preference	voted
S1	19	3.5	Dog	Yes
S2	21	3.7	Cat	No
S3	20	3.2	None	Yes
S4	22	3.8	Dog	No
S5	19	3.1	Cat	Yes

student_id	age	gpa	pet_preference	prefers_dogs	prefers_cats	prefers_none	voted
S1	19	3.5	Dog	1	0	0	Yes
S2	21	3.7	Cat	0	1	0	No
S3	20	3.2	None	0	0	1	Yes
S4	22	3.8	Dog	1	0	0	No
S5	19	3.1	Cat	0	1	0	Yes



Encoding Schemes

- Example of encoding schemes:
 - **Scheme 1: Simple enumeration** (Dislike: 1, Neutral: 2, Like: 3).
 - **Scheme 2: Centered around Neutral** (Dislike: -1, Neutral: 0, Like: 1).
 - **Scheme 3: Boolean encoding** (Dislike: 0, Neutral: 0, Like: 1).
 - **Scheme 4: Emphasizes "Like"** (Dislike: 0, Neutral: 1, Like: 1000).
- The choice of encoding scheme affects visualizations and analyses.
- The best encoding scheme depends on the type of visualization and analysis intended.

Condense the Categories of a Categorical Variable

Condensing the categories is the process of reducing the number of distinct values in a categorical variable by merging categories into broader ones. This simplifies the variable for analysis while reducing its informational detail.

Response	Encoding Scheme 1	Encoding Scheme 2	Encoding Scheme 3	Encoding Scheme 4
Dislike	1	-1	0	0
Neutral	2	0	0	1
Like	3	1	1	1000



Splitting a Variable into Multiple Variables

- Data can come in various formats, including variables with different types of responses.
- Free-response questions may result in diverse formats for the same type of information, such as birthdates.
- For analysis and visualization, it may be necessary to extract specific parts of a variable, like the month and year from a birthdate.



Splitting a Variable into Multiple Variables

- Software tools provide methods to extract and split information from categorical variables.
- Splitting a categorical variable involves creating new variables from its values, such as extracting the month and year from a date.
- **Example:** A dataset with a date variable can be split into separate month and year variables, as shown in the table:
 - March 4th, 2003 -> March, 2003
 - September 20th, 2012 -> September, 2012



Reshaping a Dataset

- Reshaping a dataset involves transforming its structure.
- **Reshaping a dataset** involves transforming its structure from a wide format, with extensive columns, to a long format, with fewer columns but more rows, or vice versa, to facilitate more effective visualization and analysis.
- A tidy dataset has each column representing a variable, each row representing an observation, and each cell containing a value.



Reshaping a Dataset

- The three variables in this dataset are country, year, and population, but the years are column names rather than values in a column.
- To tidy this dataset, it should be converted to a long format with three columns: country, year, and population, and each row representing a combination of these variables.



Reshaping a Dataset

- The long format is more suitable for computing systems for visualization and analysis, even though it may be less readable for humans.

country	year	population
Afghanistan	1800	3.28M
Afghanistan	1801	3.28M
Afghanistan	1802	3.28M
Afghanistan	1803	3.28M
Afghanistan	1804	3.28M
Afghanistan	1805	3.28M
Afghanistan	1806	3.28M
Afghanistan	1807	3.28M
Afghanistan	1808	3.28M
Afghanistan	1809	3.28M
Afghanistan	1810	3.28M
Afghanistan	1811	3.28M
Afghanistan	1812	3.28M
Afghanistan	1813	3.28M
Afghanistan	1814	3.28M
Afghanistan	1815	3.28M
Afghanistan	1816	3.28M
Afghanistan	1817	3.28M
Afghanistan	1818	3.28M
Afghanistan	1819	3.28M
Afghanistan	1820	3.29M
Afghanistan	1821	3.3M
Afghanistan	1822	3.31M
Afghanistan	1823	3.32M
Afghanistan	1824	3.34M

Reshaping a Dataset

country	1800	1801	1802	1803	1804	1805	1806	1807	1808	1809	1810
Afghanistan	3.28M	3.28M	3.28M	3.28M	3.28M	3.28M	3.28M	3.28M	3.28M	3.28M	3.28M
Angola	1.57M	1.57M	1.57M	1.57M	1.57M	1.57M	1.57M	1.57M	1.57M	1.57M	1.57M
Albania	400k	402k	404k	405k	407k	409k	411k	413k	414k	416k	418k
Andorra	2650	2650	2650	2650	2650	2650	2650	2650	2650	2650	2650
UAE	40.2k	40.2k	40.2k	40.2k	40.2k	40.2k	40.2k	40.2k	40.2k	40.2k	40.2k
Argentina	534k	520k	506k	492k	479k	466k	453k	441k	429k	417k	420k
Armenia	413k	413k	413k	413k	413k	413k	413k	413k	413k	413k	413k
Antigua and Barbuda	37k	37k	37k	37k	37k	37k	37k	37k	37k	37k	37k
Australia	200k	205k	211k	216k	222k	227k	233k	239k	246k	252k	259k
Austria	3M	3.02M	3.04M	3.05M	3.07M	3.09M	3.11M	3.12M	3.14M	3.16M	3.18M
Azerbaijan	880k	880k	880k	880k	880k	880k	880k	880k	880k	880k	880k
Burundi	899k	899k	899k	899k	899k	899k	899k	899k	899k	899k	899k
Belgium	3.25M	3.26M	3.27M	3.28M	3.29M	3.3M	3.3M	3.31M	3.32M	3.33M	3.34M

- The wide format can be more readable for humans, especially for viewing as labels on a graph.

Section 4

Data Integration

Combining Datasets

- Data science projects often involve working with a single large dataset or integrating multiple datasets.
- Combining datasets can help reveal patterns or relationships in the original data.
- Integration of multiple datasets can be challenging but is often beneficial.
- Additional cleaning and transforming may be necessary after datasets are integrated.



Stacking Datasets

- **Stacking datasets** involves combining two datasets with identical variables but different observations by placing one dataset on top of the other.
- The order of stacking does not affect the outcome.
- Example: Combining datasets with average annual rainfall and temperature for different countries.



Stacking Datasets

country	average_rainfall	average_temperature
Kenya	24.8	71.6
Tanzania	42.2	78.8
Uganda	47.2	71.6
Ethiopia	33.4	70.2

country	average_rainfall	average_temperature
China	25.4	57.2
Japan	66.0	60.8
India	42.6	75.2

country	average_rainfall	average_temperature
Kenya	24.8	71.6
Tanzania	42.2	78.8
Uganda	47.2	71.6
Ethiopia	33.4	70.2
China	25.4	57.2
Japan	66.0	60.8
India	42.6	75.2



Augmenting a Dataset

- Augmenting two datasets involves combining datasets with different variables but identical observations by placing them side by side horizontally.
- The order of combining does not matter, but the observations in both datasets must match in sequence.
- Example: Combining Japan's average rainfall and average temperature for four seasons.



Augmenting a Dataset

season	average_rainfall
Fall	16.5
Winter	16.7
Spring	9.7
Summer	23.1

season	average_temperature
Fall	59.4
Winter	43.2
Spring	58.6
Summer	81.9

season	average_rainfall	average_temperature
Fall	16.5	59.4
Winter	16.7	43.2
Spring	9.7	58.6
Summer	23.1	81.9



■ Merging Datasets

- **Merging datasets** involves combining two datasets based on at least one common variable.
- The merged dataset can be used to investigate relationships between club participation and academic performance.
- Ensure consistency in units, scales, and formats when integrating data from different sources.

■ Merging Datasets

student_id	club_name
S1	Drama Club
S1	Math Club
S2	Debate Team
S3	Math Club
S3	Science Society
S4	Drama Club
S5	Debate Team
S5	Science Society

student_id	major	gpa
S1	Math	3.8
S2	English	3.5
S3	Science	3.9
S4	Drama	3.2
S5	Politics	3.7

student_id	club_name	major	gpa
S1	Drama Club	Math	3.8
S1	Math Club	Math	3.8
S2	Debate Team	English	3.5
S3	Math Club	Science	3.9
S3	Science Society	Science	3.9
S4	Drama Club	Drama	3.2
S5	Debate Team	Politics	3.7
S5	Science Society	Politics	3.7



Final

Conclusions

Conclusions

- **Data wrangling procedures, emphasizing how they help get data ready for analysis and narrative.**
- **How to deal with missing data in a dataset.**
- **Treating the unusual numbers that could skew data analysis.**
- **Improving data structure and analysis, transform categorical and quantitative variables.**
- **Merging several datasets for in-depth data analysis.**



Comments & Questions

